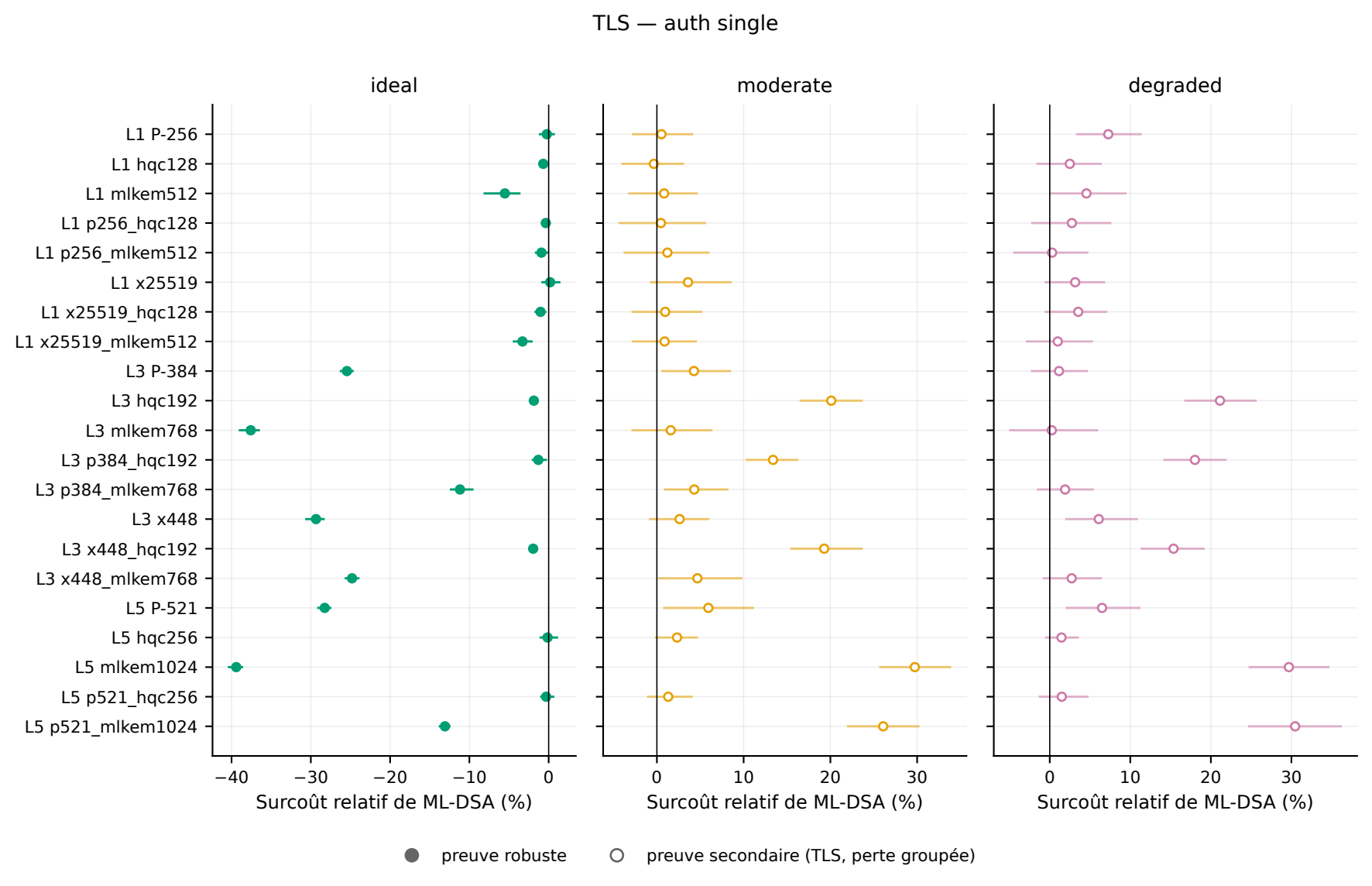
TLS — auth single
ideal
moderate
degraded
L1 P-256
L1 hqc128
L1 mlkem512
L1 p256_hqc128
L1 p256_mlkem512
L1 x25519
L1 x25519_hqc128
L1 x25519_mlkem512
L3 P-384
L3 hqc192
L3 mlkem768
L3 p384_hqc192
L3 p384_mlkem768
L3 x448
L3 x448_hqc192
L3 x448_mlkem768
L5 P-521
L5 hqc256
L5 mlkem1024
L5 p521_hqc256
L5 p521_mlkem1024
−40
−30
−20
−10
0
0
10
20
30
0
10
20
30
Surcoût relatif de ML-DSA (%)
Surcoût relatif de ML-DSA (%)
Surcoût relatif de ML-DSA (%)
preuve robuste
preuve secondaire (TLS, perte groupée)